EXACTNESS OF 0 -> PIC(T) -> PIC(X_T) -> PICARDFUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 978-987 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 989-1025, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_pic.json (pb=1895, stmt 978-987, proof 989-1025).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-flat-geometrically-connected-fibres' also exists in stacks-quot — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0B9L: sources/stacks-pic/pinned-original-source.tex, source lines 957–968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 023T: sources/stacks-pic/pinned-original-source.tex, source lines 1025–1025. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be as in Definition 0B9L. If $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism for all $T \in \mathrm{Ob}((Sch/S)_{fppf})$, then*

$$0 \to \mathrm{Pic}(T) \to \mathrm{Pic}(X_T) \to \mathrm{Pic}_{X/S}(T)$$

is an exact sequence for all T.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We may replace S by T and X by X_T and assume that $S = T$ to simplify the notation. Let $\mathcal{N}$ be an invertible $\mathcal{O}_S$-module. If $f^*\mathcal{N} \cong \mathcal{O}_X$, then we see that $f_*f^*\mathcal{N} \cong f_*\mathcal{O}_X \cong \mathcal{O}_S$ by assumption. Since $\mathcal{N}$ is locally trivial, we see that the canonical map $\mathcal{N} \to f_*f^*\mathcal{N}$ is locally an isomorphism (because $\mathcal{O}_S \to f_*f^*\mathcal{O}_S$ is an isomorphism by assumption). Hence we conclude that $\mathcal{N} \to f_*f^*\mathcal{N} \to \mathcal{O}_S$ is an isomorphism and we see that $\mathcal{N}$ is trivial. This proves the first arrow is injective.

Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module which is in the kernel of $\mathrm{Pic}(X) \to \mathrm{Pic}_{X/S}(S)$. Then there exists an fppf covering $\{S_i \to S\}$ such that $\mathcal{L}$ pulls back to the trivial invertible sheaf on X_{S_i}. Choose a trivializing section s_i. Then $\mathrm{pr}_0^*s_i$ and $\mathrm{pr}_1^*s_j$ are both trivialising sections of $\mathcal{L}$ over $X_{S_i \times_S S_j}$ and hence differ by a multiplicative unit

$$f_{ij} \in \Gamma(X_{S_i \times_S S_j}, \mathcal{O}^*_{X_{S_i \times_S S_j}}) = \Gamma(S_i \times_S S_j, \mathcal{O}^*_{S_i \times_S S_j})$$

(equality by our assumption on pushforward of structure sheaves). Of course these elements satisfy the cocycle condition on $S_i \times_S S_j \times_S S_k$, hence they define a descent datum on invertible sheaves for the fppf covering $\{S_i \to S\}$. By Descent, Proposition 023T there is an invertible $\mathcal{O}_S$-module $\mathcal{N}$ with trivializations over S_i whose associated descent datum is $\{f_{ij}\}$. Then $f^*\mathcal{N} \cong \mathcal{L}$ as the functor from descent data to modules is fully faithful (see proposition cited above). $\square$